American Computer Science League

2023-2024 ● Contest 2: Short Problems Solutions ● Classroom Division

1. Prefix/Infix/Postfix Notation

− + 2 * 3 8 / * 6 2 3 = − + 2 (* 3 8) / (* 6 2) 3
= − (+ 2 24) (/ 12 3)
= − 26 4
= 22

C. 22

2. Prefix/Infix/Postfix Notation

$$\frac{A+B}{C-1} + \frac{2*A-3}{D}$$

$= ((A\ +\ B)\ /(\ C\ -\ 1))\ +\ ((2\ *\ A\ -\ 3)\ /\ D)$
$= ((A\ B\ +)\ /\ (C\ 1\ -))\ +\ ((2\ A\ *\ 3\ -)\ /\ D)$
$= (A\ B\ +\ C\ 1\ -\ /)\ +\ (2\ A\ *\ 3\ -\ D\ /)$
$= A\ B\ +\ C\ 1\ -\ /2\ A\ *\ 3\ -\ D\ /\ +$

B. AB+C1−/2A*3−D/+

3. Bit-String Flicking

00110 OR 01100 XOR 11010
= 00110 OR (01100 XOR 11010)
= 00110 OR 10110
= 10110

E. 10110

4. Bit-String Flicking

(LSHIFT-1 01110) AND (RCIRC-2 (10110 OR NOT 10101))
= 11100 AND (RCIRC-2 (10110 OR 01010))
= 11100 AND (RCIRC-2 11110)
= 11100 AND 10111
= 10100

A. 10100

5. LISP

(CAR (CDR (CAR (CDR '(5 (2 (3 4) 1) (6 7 8) 9)))))
= (CAR (CDR (CAR '((2 (3 4) 1) (6 7 8) 9))))
= (CAR (CDR '(2 (3 4) 1)))
= (CAR '((3 4) 1))
= (3 4)

C. (3 4)

6. Prefix/Infix/Postfix Notation

5 8 − 6 9 6 − * 3 / + = (5 8 −) 6 (9 6 −) * 3 / +
= (-3) (6 3 *) 3 / +
= (-3) (18 3 /) +
= (-3) 6 +
= 3

C. 3

7. Prefix/Infix/Postfix Notation

$$\frac{(a+b)^2-c}{d+e(b^2-a)}$$

= ((a + b) ^ 2 − c)) / (d + e * (b ^2 − a))
= ((+ a b)^2 − c)) / (d + e*(^ b 2 − a))
= ((^ + a b 2) − c) / (d + e*(− ^ b 2 a))
= (− ^ + a b 2 c) / (d +(* e − ^ b 2 a))
= (− ^ + a b 2 c) / (+ d * e − ^ b 2 a)
= / − ^ + a b 2 c + d * e − ^ b 2 a

E. /−^+ab2c+d*e−^b2a

8. Bit-String Flicking

(LCIRC-2 (RSHIFT-1 (NOT 10100))) OR (RCIRC-1 (LSHIFT-2 10111))
= (LCIRC-2 (RSHIFT-1 01011)) OR (RCIRC-1 11100)
= (LCIRC-2 00101) OR 01110
= 10100 OR 01110
= 11110

D. 11110

9. Bit-String Flicking

Let X = abcde
LHS = (LCIRC-2 01101) AND X OR (RSHIFT-1 01111)
= (LCIRC-2 01101) AND abcde OR (RSHIFT-1 01111)
= (10101 AND abcde) OR 00111
= a0c0e OR 00111
= a0111
RHS = 10111
LHS = RHS ⇒ a0111 = 00111 ⇒ a = 1, b = *, c = *, d = *, e = *
Therefore x = 1****

B. 1****

10. LISP

(SETQ X '(1 2 (a (b) c) ((1 a) (2 b) c)))
(CDADDR '(1 2 (a (b) c) ((1 a) (2 b) c)))
= (CDADR '(2 (a (b) c) ((1 a) (2 b) c)))
= (CDAR '((a (b) c) ((1 a) (2 b) c)))
= (CDR '(a (b) c))
= ((b) c)

C. ((b) c)